Production Model: Diagnosis & Extension

Date: 2026-03-25

1. Current best-fitted model

The model reported in the paper is the **incremental, context-updating (recursive) speaker** with hierarchical participant effects. It won the 2×2 model comparison (speaker architecture × semantic regime):

Model	Speaker	Semantics	ELPD LOO	Δ ELPD	dSE	Δ/dSE	*p*
Inc. recursive	Incremental	Context-updating	**-7180**	0	—	—	—
Inc. static	Incremental	Context-fixed	-7199	-19	1.84	-10.28	< .001
Global static	Global	Context-fixed	-7626	-446	40.27	-11.08	< .001
Global recursive	Global	Context-updating	-7628	-448	40.69	-11.02	< .001

Key takeaways from the 2×2:

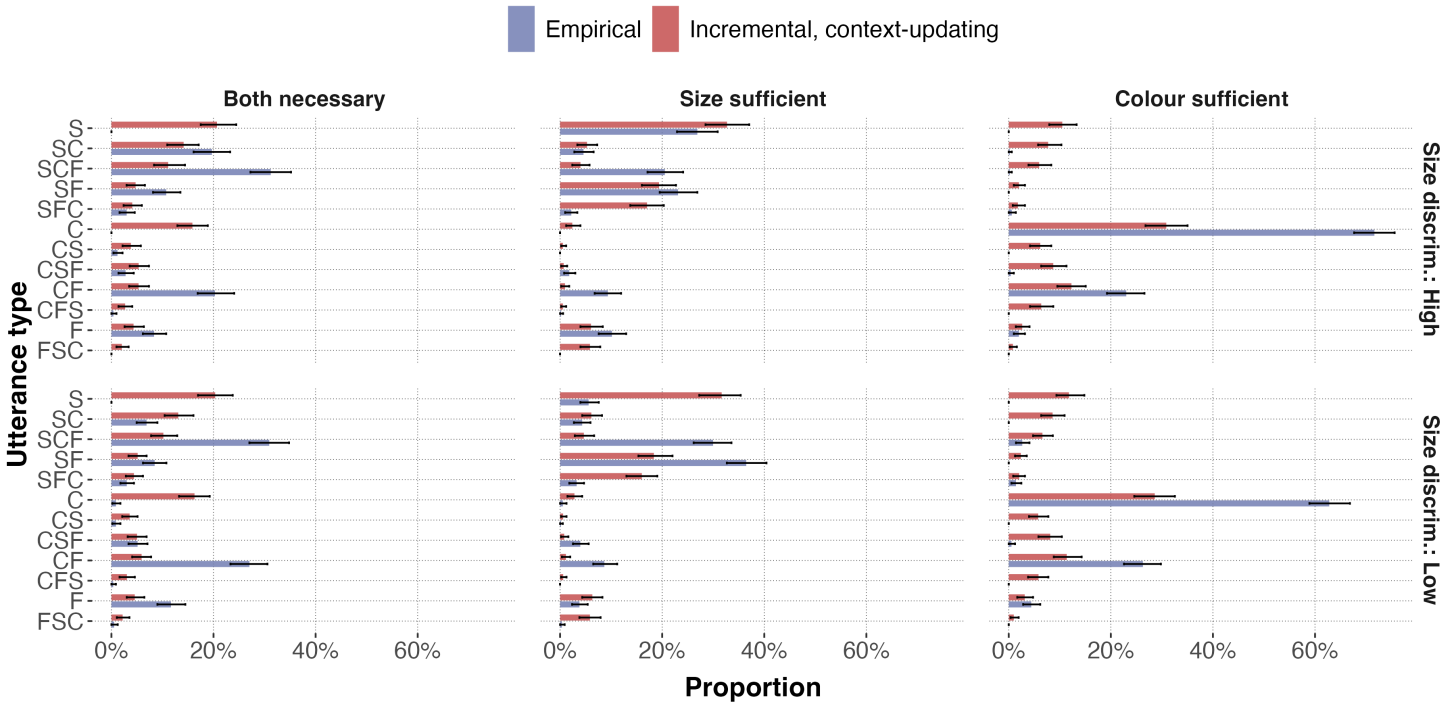
- **Speaker architecture dominates**: Incremental >> Global (Δ/dSE ≈ -11). The step-level decision structure is essential for production data.
- **Semantic regime contributes modestly**: Context-updating beats context-fixed by Δ ELPD = -19 (incremental; Δ/dSE = -10.3), a statistically reliable but small advantage.

Parameters inferred: `alpha` (rationality), `log_beta` (LM cost temperature), plus hierarchical `tau` and per-participant offsets.

For comparison, the slider model achieves $R^2 = 0.92$ on condition-level means. The production model achieves **$R^2 = 0.35$** — a much worse fit that motivates the diagnostic work below.

2. Problems visible in the current model's predictions

PPC barplot: empirical vs model predictions



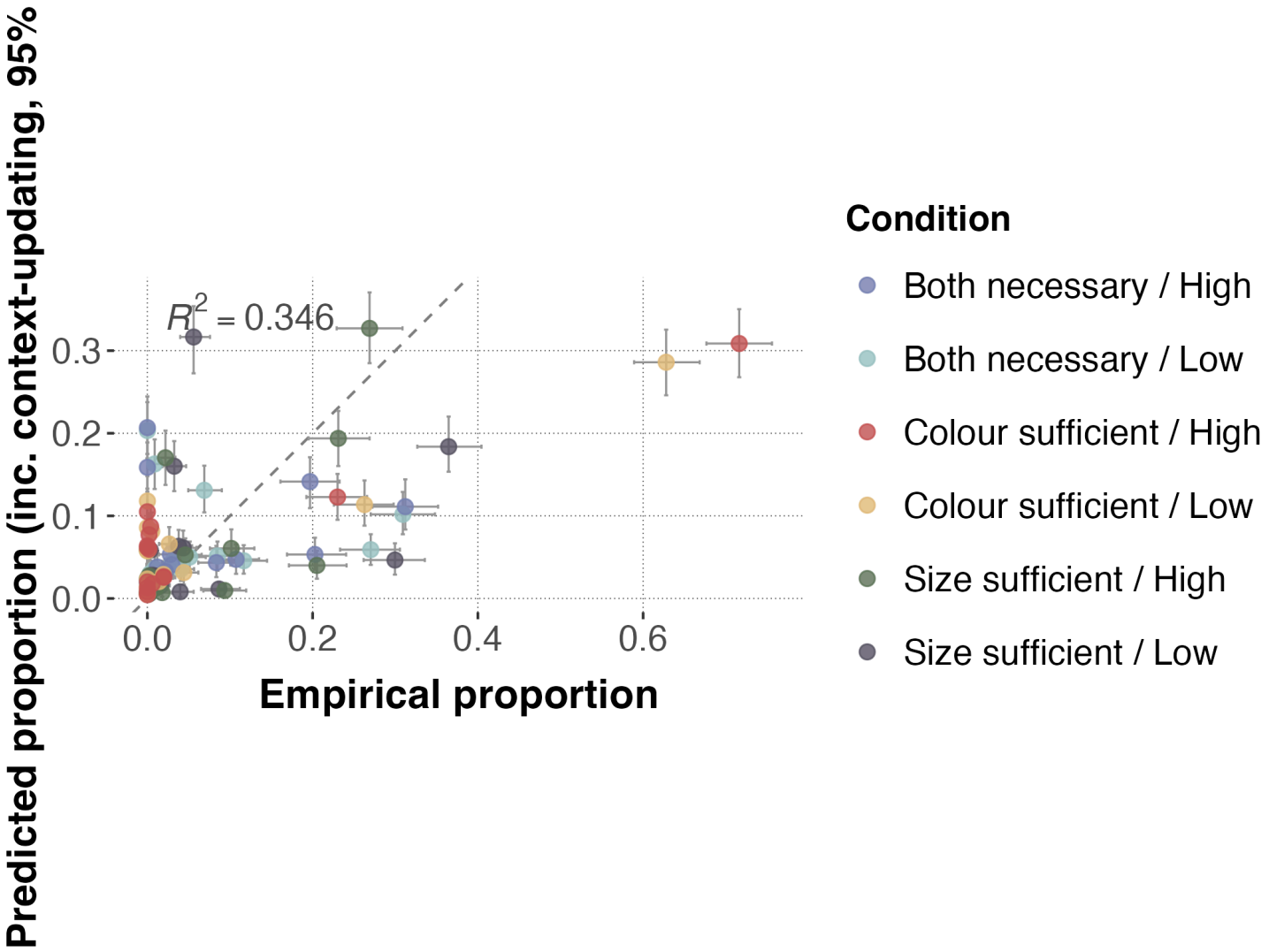
Side-by-side comparison of empirical utterance-type proportions (blue) and posterior predictive means (red) with 95% CIs, from the incremental context-updating model. Rows = size discriminability; columns = referential context.

The model captures the qualitative pattern — S-initial dominance when size is sufficient, C dominance when colour is sufficient — but four systematic misfits stand out:

1. **Overestimates bare S (size-only)** — particularly in size-sufficient | low discriminability (human 6% vs model 32%) and across both-necessary conditions, where humans rarely produce S alone but the model assigns substantial mass to it
2. **Underestimates bare C in colour-sufficient conditions** — (e.g., human 72% vs model 31% in colour-sufficient | high discriminability)
3. **Underestimates CF in both-necessary conditions** — CF is the most frequent utterance type in both-necessary (human ~25% in low discriminability), but the model substantially underpredicts it
4. **Underestimates three-adjective (overspecified) utterances** — (e.g., human 31% vs model 11% for SCF in both-necessary | high discriminability)
5. **Sparse predictions vs degenerate data in colour-sufficient** — empirically, colour-sufficient conditions are dominated by bare C (60–72%), with most other types near zero. The model

instead spreads mass across many utterance types, failing to capture the near-degenerate distribution where one category accounts for the vast majority of responses

Correlation: predicted vs empirical proportions



Each point is one utterance type in one condition (90 points total). The model's predictions cluster closer to the centre than the data — it cannot concentrate enough probability mass on the dominant utterance in each condition, and overestimates rare types. $R^2 = .35$ ($r = .59$, $p < .001$, $n = 90$), but this is inflated by the many near-zero utterance types where both model and data agree on ~0. Filtering to types with ≥ 2% proportion gives $R^2 = .30$ ($r = .55$, $p < .001$, $n = 70$); filtering to ≥ 5% gives $R^2 = .21$ ($r = .46$, $p < .001$, $n = 48$). The correlation remains reliable but weakens as the easy-to-predict near-zero categories are removed, indicating room for improvement on the behaviourally relevant utterance types.

3. Why the current model struggles

The misfits trace to structural limitations of the original model:

Insufficient utility differentiation across conditions

The experimental conditions create different referential contexts (which properties are necessary, how discriminable size is), and RSA translates these into different relative utility profiles over the 15 utterance types. Alpha amplifies these utility differences via softmax: a larger alpha means the model concentrates more mass on the highest-utility utterance in each condition. The core problem is that the utility differences the model computes are **not large enough** to match the sharp condition-specific preferences humans show. Two factors contribute:

1. **The model is not sensitive enough to dimensional differences.** A single alpha governs all dimensions equally. But size is gradable and context-dependent (perceptual similarity varies with discriminability), while colour and form are near-deterministic (blue/not-blue, circle/not-circle). Previous work on overinformativity (e.g., Pechmann 1989; Koolen et al. 2011; Rubio-Fernández 2016; Degen et al. 2020) has documented the asymmetry in redundant mention: speakers freely add colour when it is not strictly necessary but avoid adding size redundantly. The Degen et al. RSA model can already account for this asymmetry through relaxed semantics and a single alpha. However, our production data suggests that participants may employ **qualitatively different strategies** — some engage in full utility-based reasoning (deliberating over informativeness at each step), while others adopt a simpler mention-all or mention-salient strategy (e.g., always include colour because it is easy to encode). These strategies would require different alpha values, and a single alpha must compromise between them. This points toward a **mixture model** with strategy-specific rationality parameters, rather than a single-alpha model that tries to average across heterogeneous response strategies.
2. **The input representations may not be discriminating enough.** The semantic values fed to the model (continuous values between 0 and 1 for each dimension) determine the raw utility landscape before alpha scales it. If the representation does not sharply differentiate the target from distractors on the relevant dimension — e.g., if colour semantics are too close to 0.5 instead of near 1.0 — then even a high alpha cannot produce the peaked distributions that humans show. The model's relaxed semantics are a strength for enabling overspecification (cf. Degen et al. 2020), but if the semantic values understate the true perceptual discriminability, the utility differences remain too compressed.

The consequence: bare S is overproduced because the model overestimates its relative utility (treating size informativeness on par with colour), and bare C is underproduced because the model

underestimates the dominance of colour in colour-sufficient contexts.

No explicit length bias, and implicit overspecification is insufficient

The model does have an implicit overspecification mechanism: relaxed (continuous) semantics means longer utterances like SCF can have higher utility than shorter ones like S when the extra adjectives provide non-zero informativeness — this follows the approach of Degen et al. (2020). However, there is no explicit length bias to boost multi-word utterances. If anything, the LM prior penalises longer utterances (each additional word reduces probability). This interacts problematically with alpha: a high alpha sharpens the distribution toward the single highest-utility utterance, producing near-degenerate predictions, while a low alpha spreads mass too evenly. The model must simultaneously fit conditions where different utterance types dominate (e.g., push C high in colour-sufficient, push SCF high in both-necessary), but a single alpha cannot achieve condition-specific concentration without an independent length preference.

No explicit response noise

The softmax over alpha-weighted utilities does introduce implicit stochasticity — with finite alpha, all utterances receive some probability. But there is no dedicated noise component (analogous to sigma in the slider model's ZOIB likelihood) to absorb responses that are uninformative or seemingly random. When humans occasionally produce utterances like F-only in a size-sufficient context, the model can only accommodate them by lowering alpha globally, which flattens the entire distribution and pulls mass away from dominant utterances.

Net effect: flattened predictions from a three-way tension

These factors interact through alpha, which must simultaneously serve three roles: (a) amplify utility differences across conditions, (b) implicitly accommodate strategy heterogeneity (some participants reason carefully, others mention-all), and (c) absorb response noise in the absence of an explicit noise floor. A high alpha sharpens predictions toward one utterance per condition but cannot accommodate the heterogeneous strategies or rare responses; a low alpha spreads mass to cover these but washes out the condition-specific concentration. Without per-dimension rationality, an explicit length bias, or an explicit noise component, the model settles on an intermediate alpha that compromises in all three directions — producing predictions that are too flat relative to the peaked, condition-specific distributions humans actually show. The input representations compound this: if semantic values do not sharply differentiate the target on the relevant dimension, even the right alpha cannot recover the needed utility contrasts.

4. Extension attempts

Each extension targets one or more of the three tensions identified above. All extensions build on the incremental recursive speaker and are **not yet reported** in the paper.

4.1 Per-dimension alpha + length bias + lapse rate (v1 — current best, but not the one reported in the draft)

This extension directly addresses all three tensions:

- **Insufficient sensitivity** → per-dimension alpha (`alpha_D` , `alpha_C` , `alpha_F`), allowing the model to weight size deliberation differently from colour/form
- **No explicit length bias** → `gamma` , a per-extra-word bonus that explicitly rewards overspecification
- **No explicit noise** → `epsilon` , a lapse rate mixing uniform noise into the likelihood

Mechanism	Parameter	Prior	Posterior mean	Rationale
Per-dim rationality	`alpha_D`	HalfNormal(5)	6.94	Size weighted ~3.5× more than colour/form
	`alpha_C`	HalfNormal(5)	1.84	
	`alpha_F`	HalfNormal(5)	1.88	
Length bias	`gamma`	Normal(0, 1)	2.32	exp(γ) ≈ 10× bonus per extra word
Lapse rate	`epsilon`	Beta(1, 50)	0.23	~23% of responses treated as noise
LM cost temperature	`log_beta`	Normal(0, 0.5)	2.05	Sharper LM prior (β ≈ 7.8)
Form semantics	`form_sem`	fixed	0.50	Lowered from 0.841
Hierarchical	`tau` , `delta`	HalfNormal(0.2)	τ = 0.91	Per-participant offset on all three alphas

Result: R^2 = 0.596, ELPD = -6387 (+793 over original). Divergences = 0, r_hat ≤ 1.05.

Interpretation of per-dim alpha: `alpha_D = 6.94` vs `alpha_C ≈ alpha_F ≈ 1.86` — the speaker treats size informativeness ~3.5× more seriously, consistent with size being the gradable dimension that requires more deliberation. Note that this is a partial solution to the strategy heterogeneity problem: rather than a true mixture model, per-dim alpha lets the model approximate different levels of reasoning for different dimensions within a single-strategy framework.

4.2 First-word intercepts (v2) — rejected

Targeted the C-underestimation by adding utterance-level intercepts for C-first and F-first (S = reference).

- **Result**: phi_C = -6.89 (penalty, opposite of expected), phi_F = -5.90. R^2 = 0.615, ELPD = -6337
- **Rejected**: the biases went in the wrong direction. The misfit is condition-dependent — C needs a boost in colour-sufficient but a penalty in size-sufficient. Condition-independent intercepts cannot capture this, confirming that the problem is about context-sensitive utility differentiation, not a global utterance-type bias.

4.3 Step-level mention biases with per-dim alpha (v3) — rejected

Targeted representation sensitivity by adding `mu_vec` biases to incremental choice logits: `logits = alpha_vec * log_L_ref + mu_vec`.

- **Result**: n_eff ~ 2, r_hat up to 14.7 for mu_C
- **Rejected**: alpha and mu are not jointly identifiable — they trade off on the same logit scale. This is a fundamental identifiability issue, not a sampling problem.

4.4 Step-level mention biases with single alpha (v4) — rejected

Tested whether mu biases could substitute for per-dim alpha: single alpha + mu_C, mu_F.

- **Result**: R^2 = 0.540, ELPD = -6449 (worse than v1)
- **Rejected**: losing per-dim alpha hurts more than mu helps. Per-dim alpha captures something that condition-independent biases cannot — it modulates how strongly the model responds to informativeness *per dimension*, which interacts with the referential context.

5. Diagnositics of the current best model (extenden-v1)

5.1 Extended 2×2 comparison

Applied the v1 mechanisms (gamma, epsilon, form_sem = 0.50) to all 4 speaker × semantics variants to test whether the original 2×2 findings hold. Per-dim alpha for incremental only (architecturally motivated — step-level choices enable dimension-specific rationality); global models use single alpha with gamma and epsilon fixed at incremental posterior means (not identifiable in the global speaker even with 5000 warmup).

Model	Speaker	Semantics	ELPD LOO	Δ ELPD	dSE	Δ/dSE	*p*
Inc. recursive (ext.)	Incremental	Context-updating	**-6386**	0	—	—	—
Inc. recursive (ext.)	Incremental	Context-updating	-6387	-1	1.23	-0.76	.447
Inc. static (ext.)	Incremental	Context-fixed	-6392	-5	1.63	-3.25	.001
Global recursive (ext.)	Global	Context-updating	-9242	-2855	97.88	-29.17	< .001
Global static (ext.)	Global	Context-fixed	-9259	-2872	97.49	-29.46	< .001

Note: The first two rows are separate inference runs of the same model specification (incremental recursive with extended mechanisms); the Δ = -1 confirms replication stability. Global models use single alpha with gamma and epsilon fixed at incremental posterior means (not identifiable in the global speaker).

Key findings:

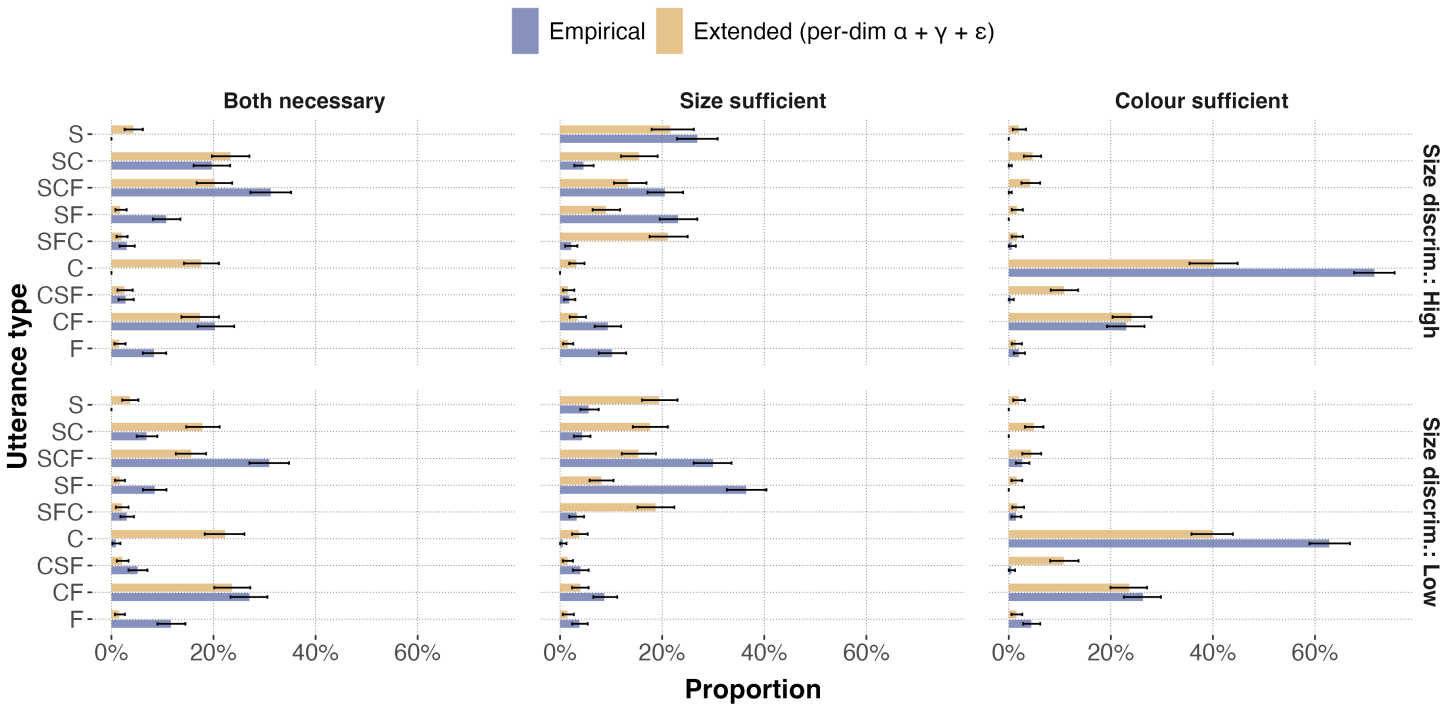
1. **Semantic regime effect survives the extensions.** Context-updating beats context-fixed by Δ ELPD = -5 (Δ/dSE = -3.25, *p* = .001). The effect is smaller than in the original 2×2 (Δ/dSE = -10.3)

because per-dim alpha and gamma absorb some of the variance, but it remains statistically significant. This is consistent with the main story in the paper: context-updating semantics provide an advantage that is not fully reducible to the other mechanisms.

2. **Speaker architecture still dominates.** Incremental >> Global (Δ/dSE ≈ -29, $p < .001$), reinforcing that step-level decision structure is the primary factor in production.

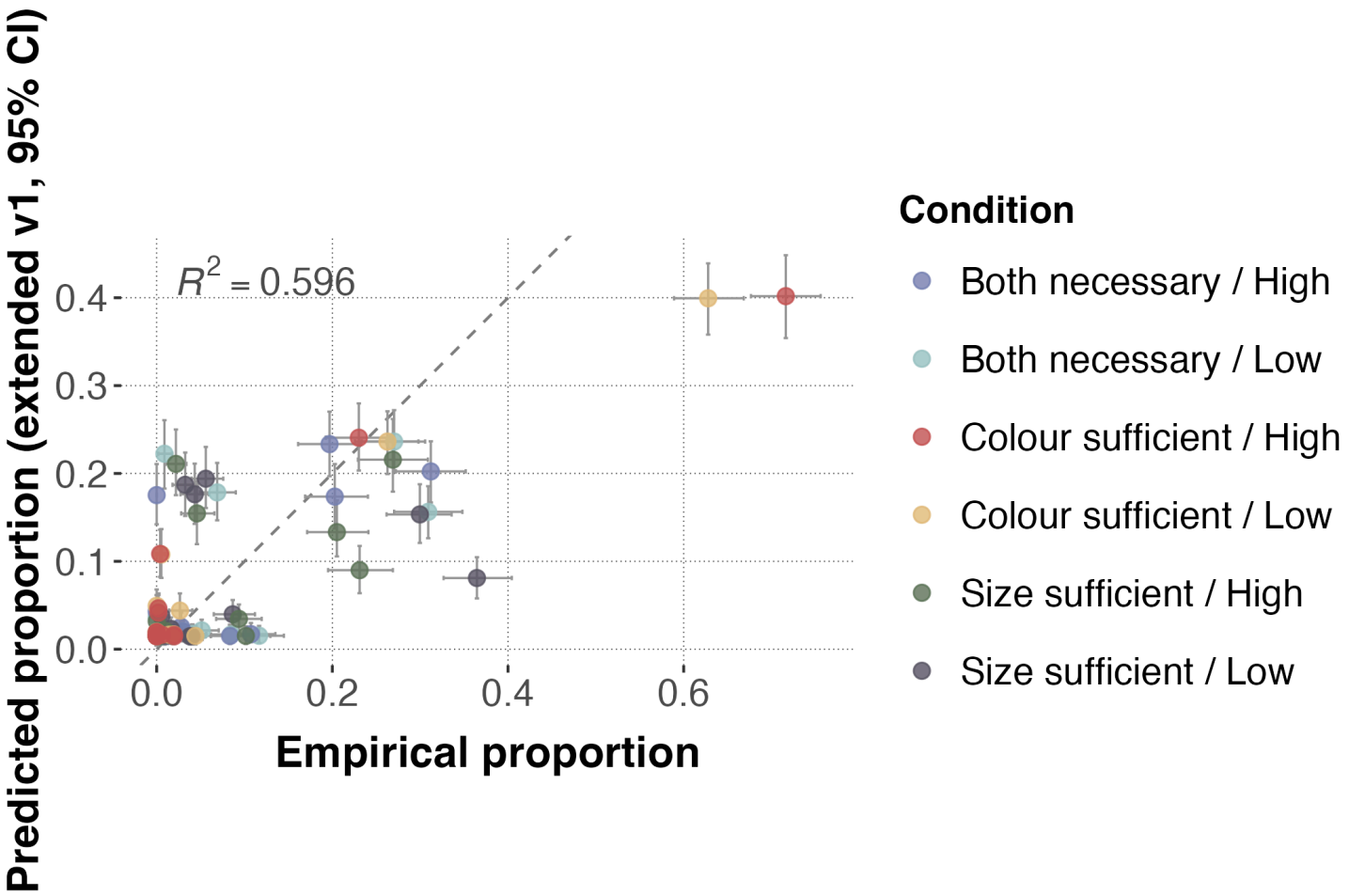
5.2. Extended model predictions

PPC barplot: empirical vs extended model



Side-by-side comparison of empirical (blue) and extended model predictions (gold) with 95% CIs. Compared to the original model (section 2), the extended model better captures the dominance of C in colour-sufficient conditions and the spread of utterance types in both-necessary. However, residual misfits remain (see below).

Correlation: predicted vs empirical (extended model)



$R^2 = .596$ ($r = .77$, $p < .001$, $n = 90$) on the full 90 points — a marked improvement over the original model's $R^2 = .346$. As with the original, this is inflated by near-zero agreement; filtering to ≥ 2% gives $R^2 = .501$ ($r = .71$, $p < .001$, $n = 50$); filtering to ≥ 5% gives $R^2 = .370$ ($r = .61$, $p < .001$, $n = 32$). The correlation remains reliable across all thresholds and is markedly improved over the original ($R^2 = .346$ / .296 / .213 at the same cuts). Points are closer to the identity line, especially for high-proportion utterance types, but the model still underestimates the most dominant categories.

5.3. Remaining misfits and structural limitations

Largest residuals:

Condition	Utterance	Human	Model	Δ
colour-sufficient | high	C	.716	.402	-.314

Condition	Utterance	Human	Model	Δ
size-sufficient | low	SF	.365	.081	-.284
colour-sufficient | low	C	.628	.399	-.228
both-necessary | low	C	.009	.222	+.214
size-sufficient | high	SFC	.022	.211	+.189
both-necessary | high	C	.000	.175	+.175
size-sufficient | low	SFC	.032	.187	+.155

Even with the extensions, the model cannot fully concentrate mass on dominant utterances. The residual pattern is informative:

- **C still underestimated in colour-sufficient**: per-dim alpha helps (`alpha_C = 1.84` vs original single alpha) but the asymmetry between colour's near-deterministic semantics and size's gradable semantics is not fully resolved by dimension-specific rationality alone. The model predicts ~40% C where humans produce 63–72%.
- **C overestimated in both-necessary**: the model assigns ~18–22% to C where humans produce 0–1%. The lapse rate (epsilon = 0.23) and the model's inability to suppress C in contexts where colour is not uniquely identifying drive this.
- **SFC overestimated in size-sufficient**: the model predicts ~19–21% for three-adjective utterances where humans produce only 2–3%. The length bias (gamma = 2.32) overcompensates in conditions where overspecification is rare.
- **Epsilon = 0.23 is uncomfortably high**: nearly 1 in 4 responses treated as uniform noise. This likely absorbs both genuine response noise *and* strategy heterogeneity that the single-strategy model cannot capture. A mixture model (utility-reasoning strategy vs mention-salient strategy) could potentially reduce epsilon by accounting for the heterogeneity structurally.

6. Open questions

1. **Should the paper report the extended model?** The extensions improve fit (Δ ELPD = +793) and each mechanism has theoretical grounding:
   - Per-dim alpha → dimension-specific salience (gradable vs deterministic features)
   - Gamma → overspecification preference (well-documented in referring expression literature)
   - Epsilon → response noise (standard in computational cognitive science)

   However, reporting the extensions changes the 2×2 narrative — the semantic regime effect weakens — and the high epsilon raises questions about model completeness.

2. **Mixture model.** One possible direction is to address strategy heterogeneity directly. A mixture of (a) a high-alpha utility-reasoning speaker and (b) a low-alpha / mention-salient speaker could replace both the per-dim alpha approximation and the high epsilon. This would provide a more principled account of why participants show such different response profiles within the same condition, and could reduce epsilon by absorbing the heterogeneity structurally rather than through a uniform noise floor.
3. **Input representations.** Another possible direction is to re-examine the semantic values fed to the model. If the continuous size semantics understate the perceptual contrast between target and distractors, no amount of alpha tuning will recover the correct utility landscape. This could be tested by fitting size semantic values as free parameters or by using empirically measured similarity ratings. The two directions are not mutually exclusive — strategy heterogeneity and representation sharpness may both contribute to the remaining misfits.

Files

- **Model code**: `05-modelling-production-data/modelSpecification.py` — `incremental_speaker_extended` and updated 4 main speaker functions
- **Inference data** (server): `inference_data/mcmc_results_*_warmup2000_samples1000_chains4.nc`
- **Figures**: `figures/production_ppc_barplot.pdf`, `figures/production_correlation_inc_hier.pdf`